OSITO DRONE STEM ACADEMY

SERIES 02: DRONE MATH: ADDITION IN THE SKY - AGES 5-7 (GRADES K-1)

THIS STEM COLORING BOOK BELONGS TO

YOUNG PILOT'S NAME / NOMBRE DEL PILOTO

MY DRONE'S CALLSIGN / NOMBRE DE MI DRON

CO-PILOT / GROWN-UP CO-PILOT NAME

Title: Vol. 052: Mateo's Dual Drone Rotor Addition at 3D Printing & Maker Workshop

Featured STEM Pilot: Mateo (Junior Aviator & Community Garden Explorer)

Flight Environment: 3D Printing & Maker Workshop - Target Age: Ages 5-7 (Grades K-1)

Core STEM & Math Focus: 4 Rotors Plus 4 Rotors Equals 8

Includes: 8 Single-Sided Coloring Plates, 4 Vector STEM Activities, Flight Log & Certificate!

Explore Youth Drone STEM Camps, 3D Printing & Flight Simulators at ositodrone.com

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OFFICIAL YOUTH DRONE SAFETY RULES & FAA TRUST NOTICE

SAFETY RULES FOR RECREATIONAL DRONE PILOTS

1

1. ALWAYS FLY WITH A GROWN-UP

Make sure a parent, teacher, or coach supervises every flight.

2

2. FLY IN OPEN, SAFE SPACES

Stay clear of people, pets, roads, trees, and active airport airspace.

3

3. KEEP DIRECT LINE-OF-SIGHT

Always keep your quadcopter where you can see it with your own eyes.

4

4. USE DUCTED PROPELLER GUARDS

Inspect all 4 ducted prop guards and battery latch before takeoff.

5

5. COUNT & CHECK PRE-FLIGHT MATH

4 propellers + 1 charged battery + 1 adult co-pilot = Safe Takeoff!

6

6. TAKE THE FREE FAA TRUST TEST

Learn official recreational sky rules at faa.gov/uas/trust with family.

OSITO DRONE LLC – PUBLISHING & LEGAL ATTRIBUTION

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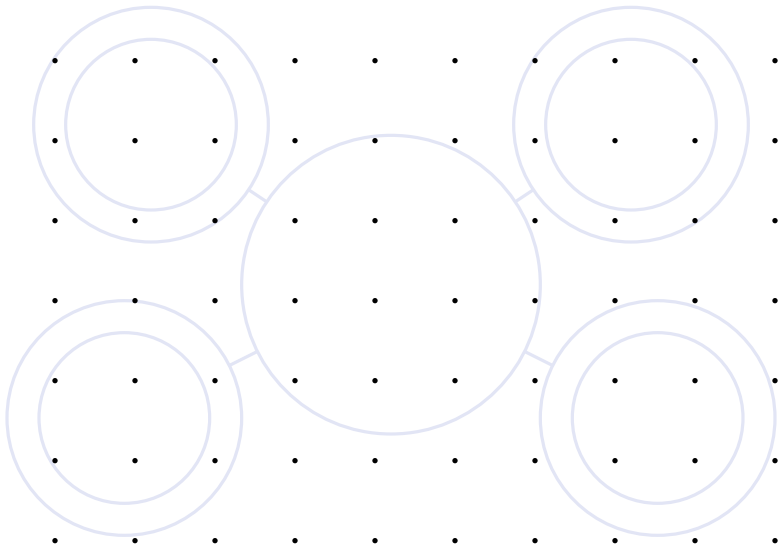
Designed to comply with FAA TRUST youth aviation educational benchmarks.

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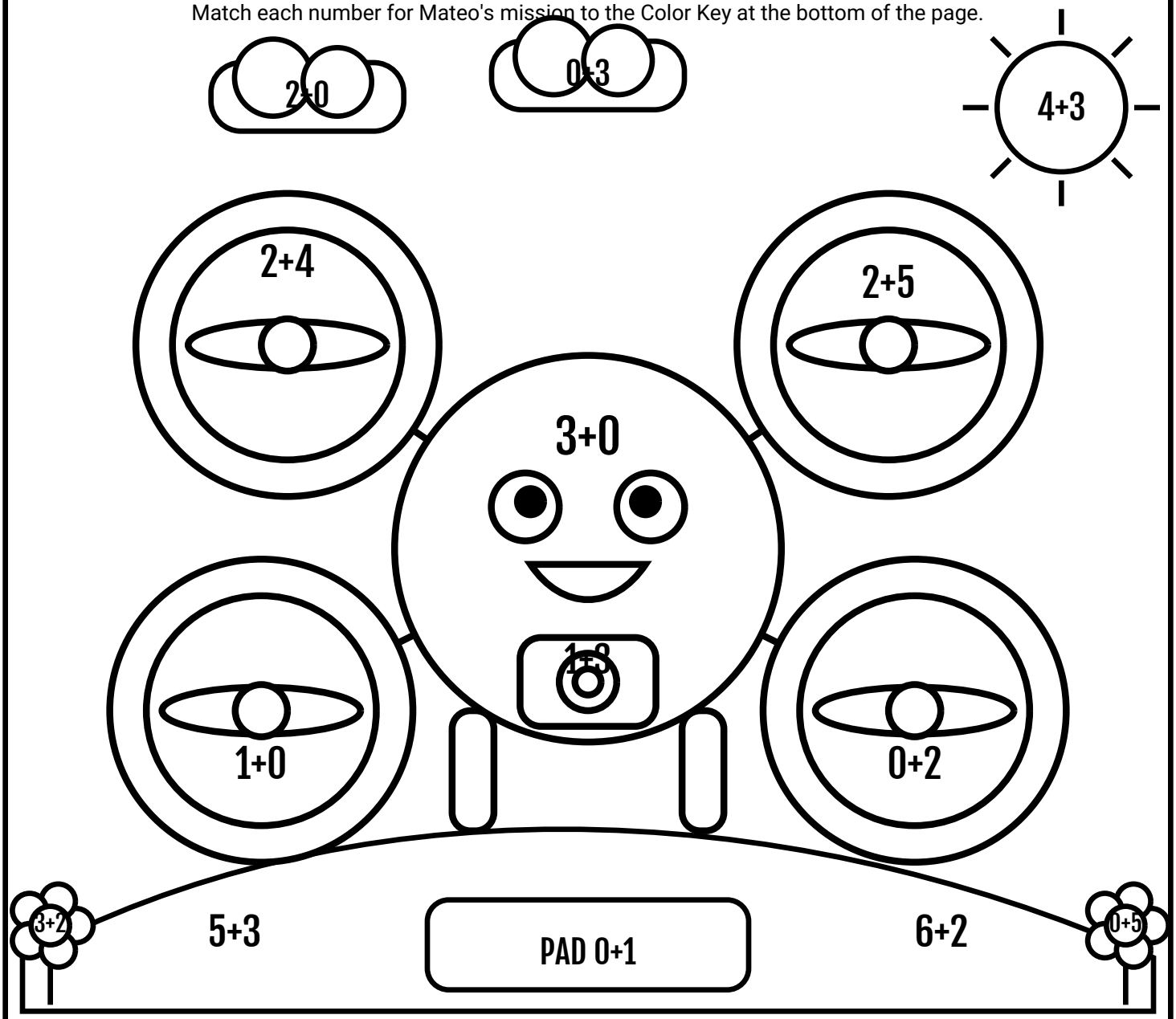


MATEO & PET DRONE – DUAL DRONE ROTOR ADDITION



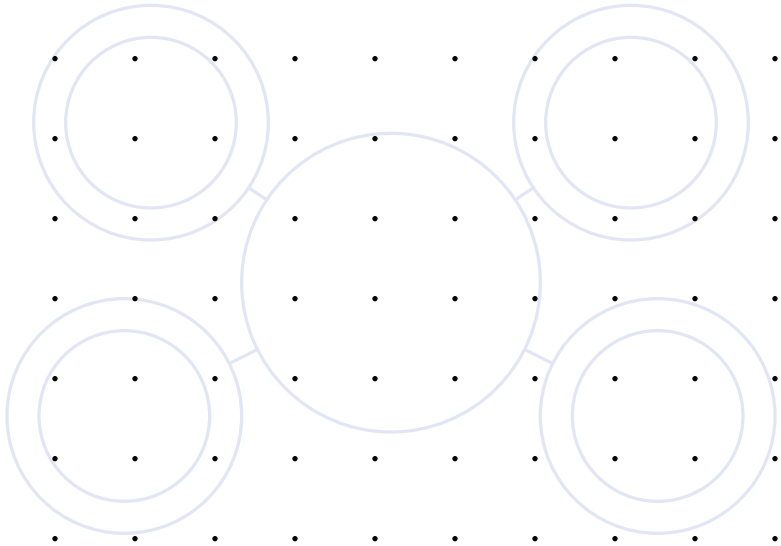
COLOR BY NUMBER: MATEO'S DUAL DRONE ROTOR ADDITION

Match each number for Mateo's mission to the Color Key at the bottom of the page.



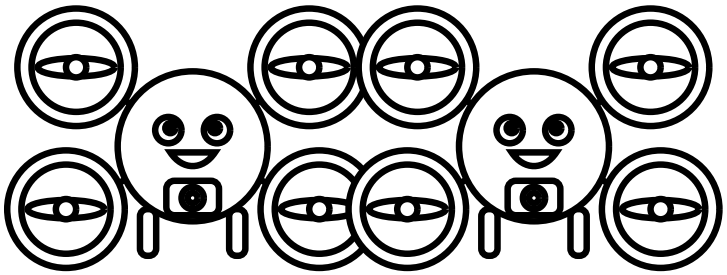
NUMBER-BY-NUMBER COLOR KEY

1 = RED	2 = BLUE	3 = YELLOW	4 = GREEN
5 = ORANGE	6 = PURPLE	7 = TEAL	8 = BROWN



DRONE MATH: MATEO'S DUAL DRONE ROTOR ADDITION

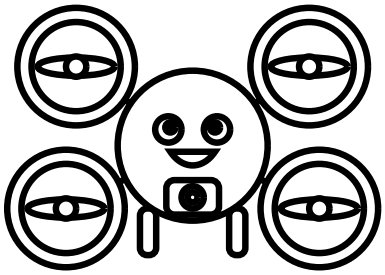
Every quadcopter has 4 ducted propellers! Solve Mateo's flight problems and trace the numbers.



MISSION #1: ADD ROTORS

8 Props + 5 = 13

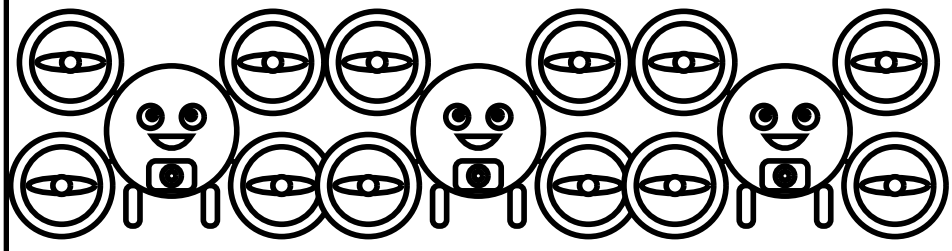
Color the drones & trace numbers!



MISSION #2: ADD ROTORS

4 Props + 6 = 10

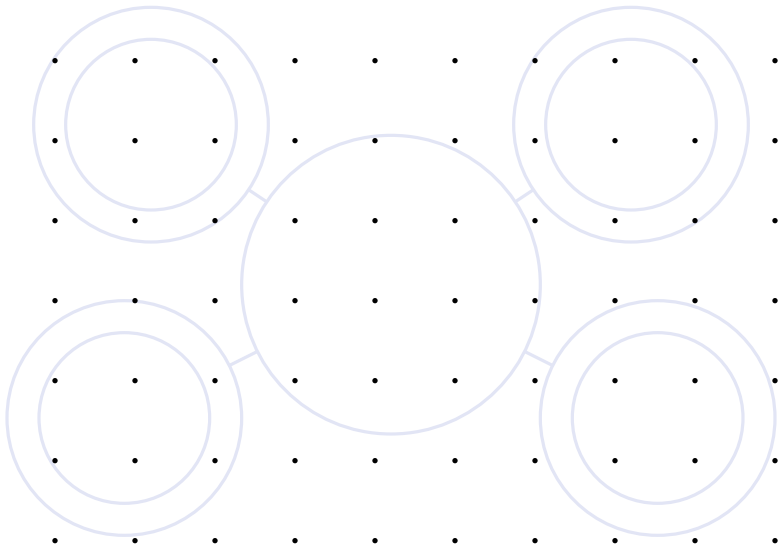
Color the drones & trace numbers!



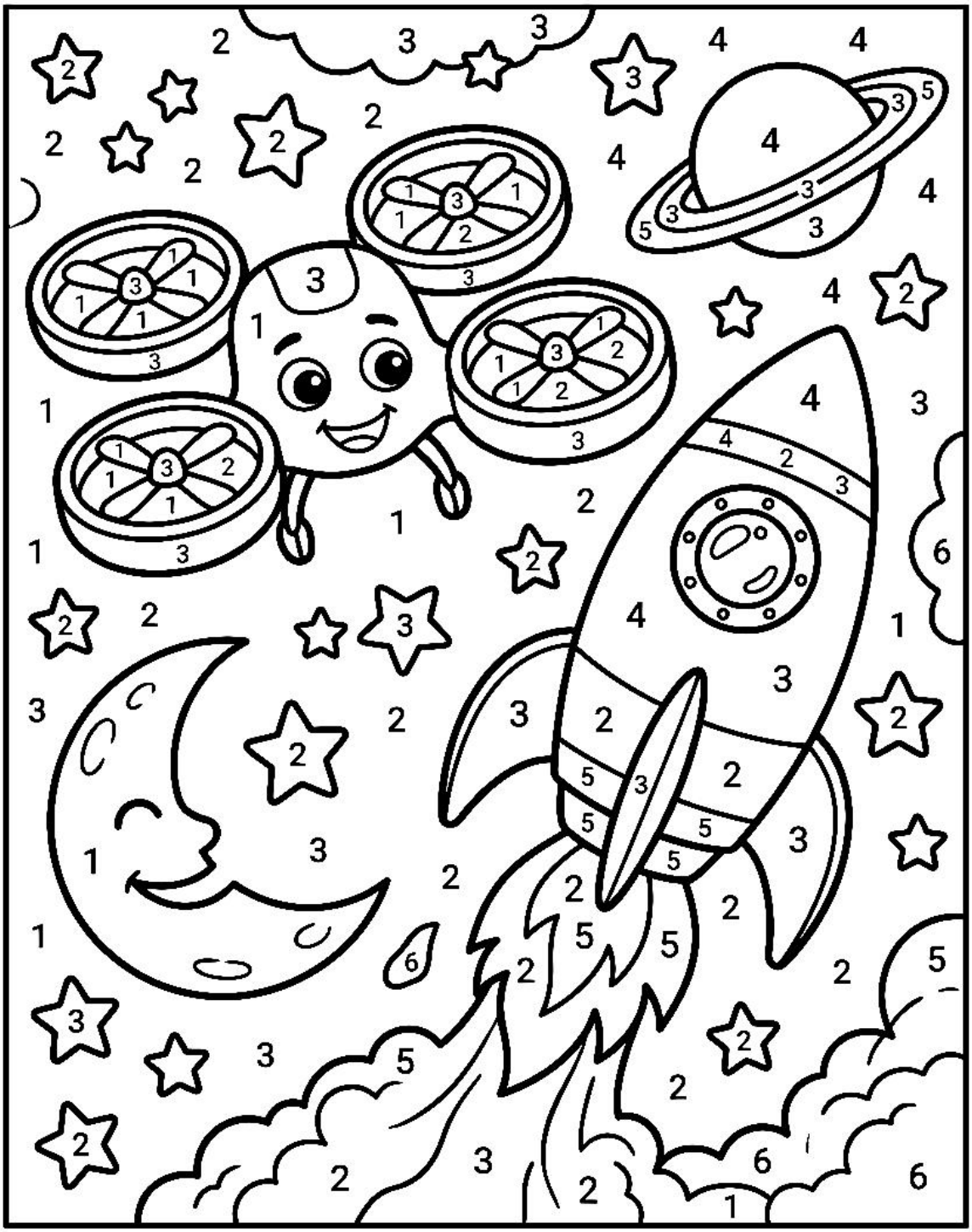
MISSION #3: ADD ROTORS

12 Props + 1 = 13

Color the drones & trace numbers!



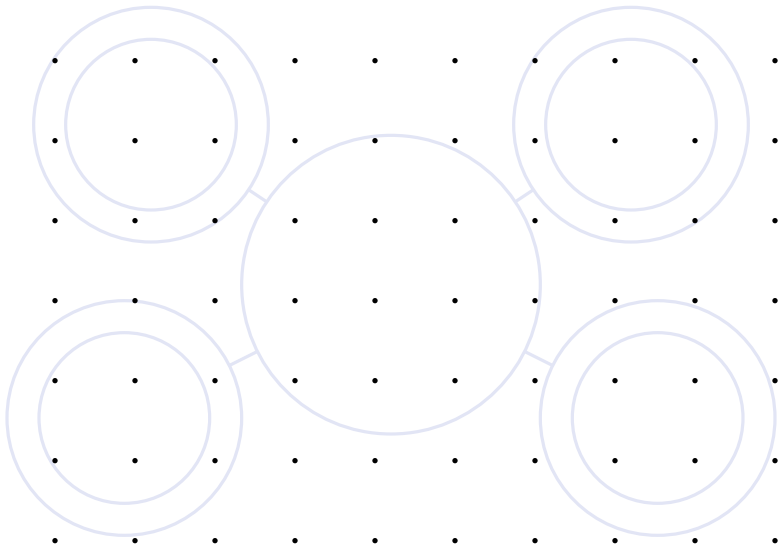
COLOR BY NUMBER: SPACE ADVENTURE



1=Blue 2=Yellow 3=Red 4=Orange 5=Green 6=Black

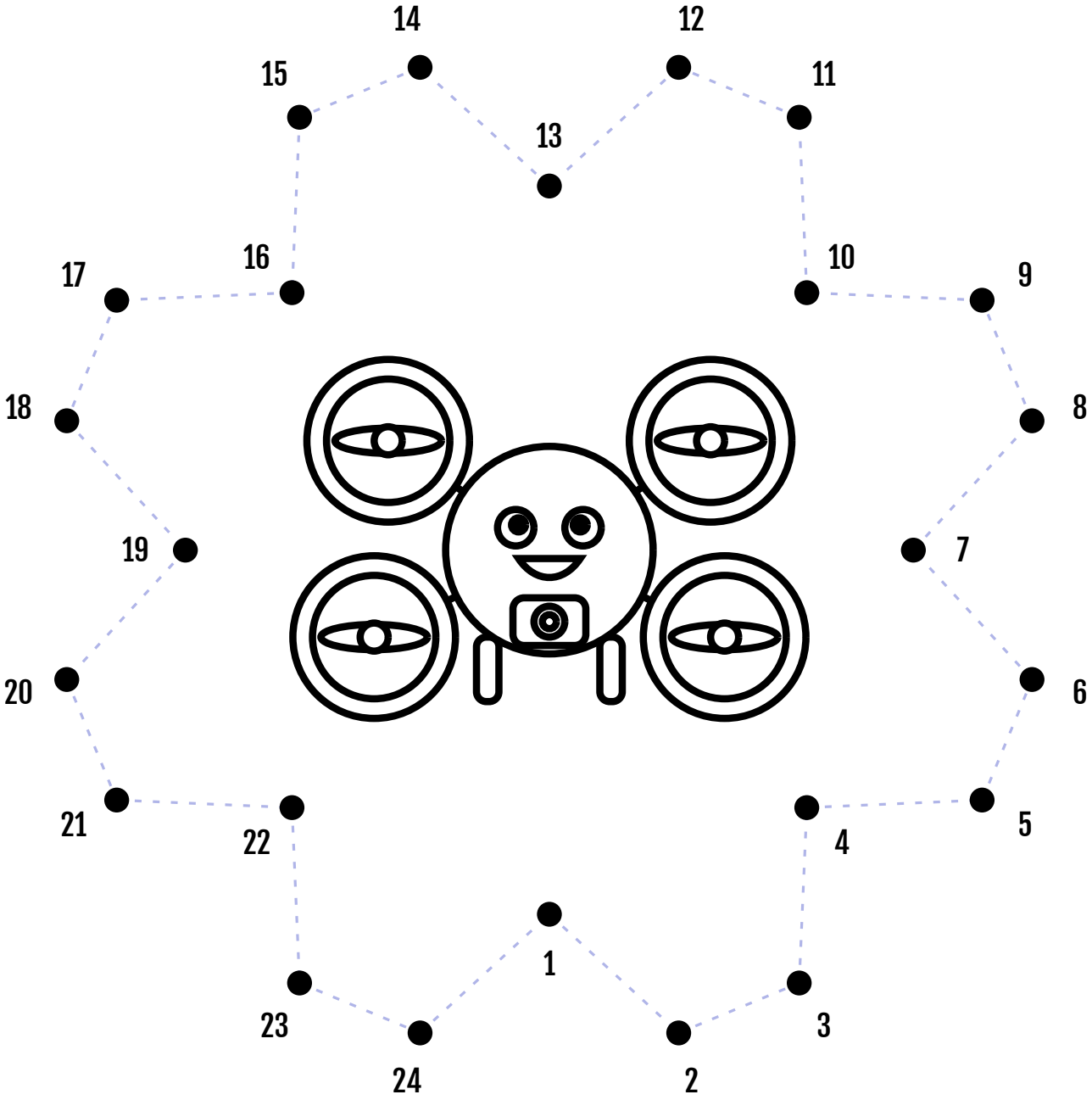
COLOR BY NUMBER & COUNT ROTORS WITH MATEO!

BLEED-PREVENTION BACKING PAGE 10 - BOOK #052 - FLIGHT LOG & SKETCH PAD



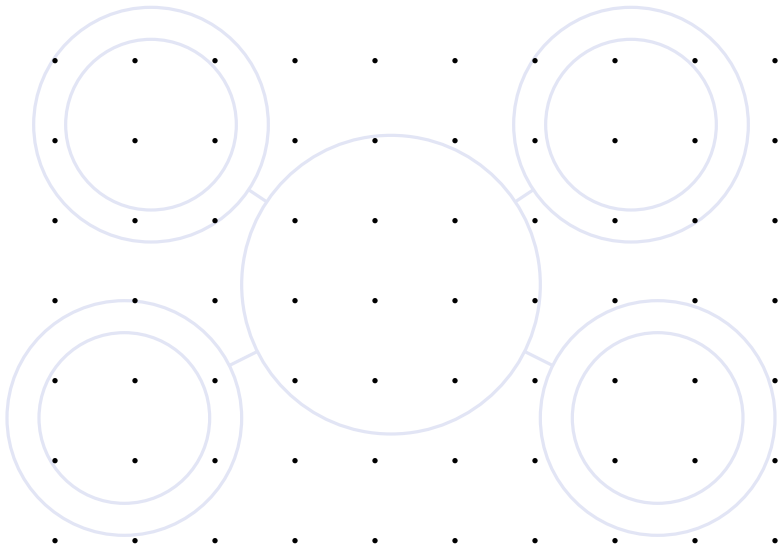
CONNECT THE DOTS: MATEO'S DUAL DRONE ROTOR ADDITION

Connect the numbered dots in order counting by 1s to complete the aerospace blueprint!



BONUS FOR MATEO: You connected 24 dots counting by 1s! Final number = 24!

BLEED-PREVENTION BACKING PAGE 12 - BOOK #052 - FLIGHT LOG & SKETCH PAD



CODE-BY-NUMBER: SOLVE & COLOR PIXEL DRONE

Solve or match the number in each grid square (A1-G7) using the Color Code Key below to reveal pixel art!

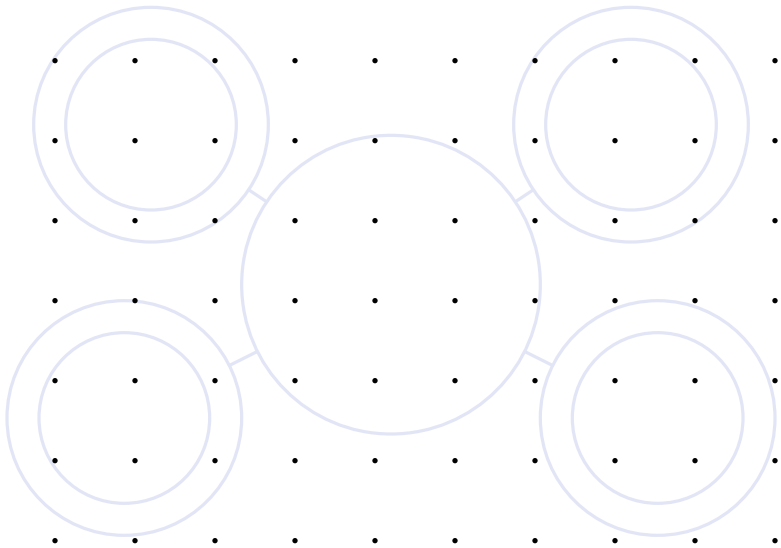
	1	2	3	4	5	6	7
A	2+0	2+0	5+0	2+3	4+1	1+1	0+2
B	1+1	1+0	0+2	3+2	1+1	1+0	0+2
C	1+4	1+1	3+0	2+1	3+0	1+1	5+0
D	3+2	2+3	0+3	0+1	0+3	2+3	2+3
E	3+2	2+0	1+2	1+2	3+0	2+0	2+3
F	1+1	0+1	1+1	0+5	1+1	0+1	1+1
G	2+0	1+1	5+0	0+5	2+3	0+2	1+1

PIXEL CODE COLOR KEY

1 = RED 2 = BLUE / INDIGO 3 = GOLD / YELLOW 4 = GREEN 5 = LIGHT SKY BLUE

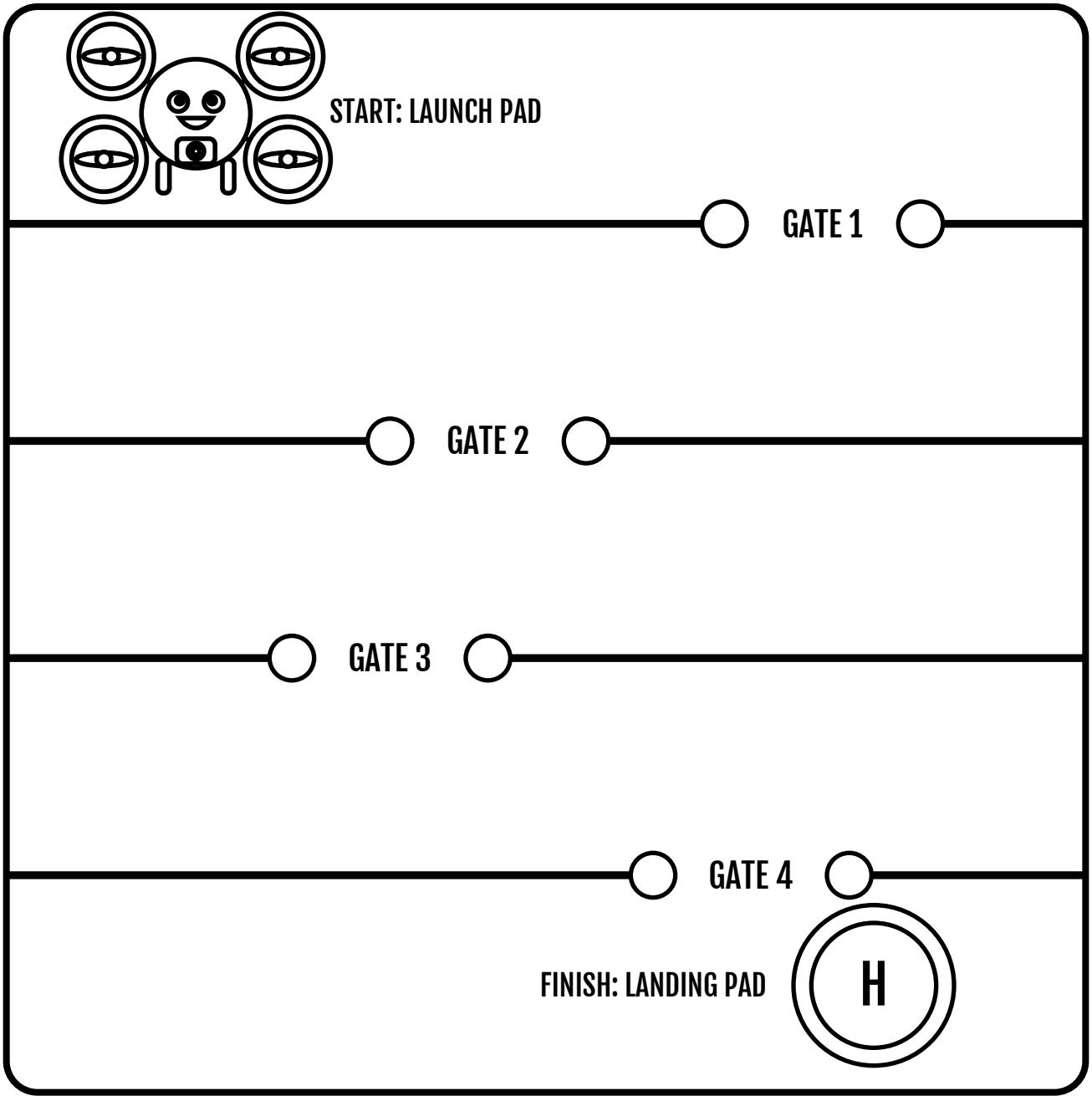
STEM Tip: Computers and drone cameras store pictures as grids of numbered pixels just like this!

BLEED-PREVENTION BACKING PAGE 14 - BOOK #052 - FLIGHT LOG & SKETCH PAD



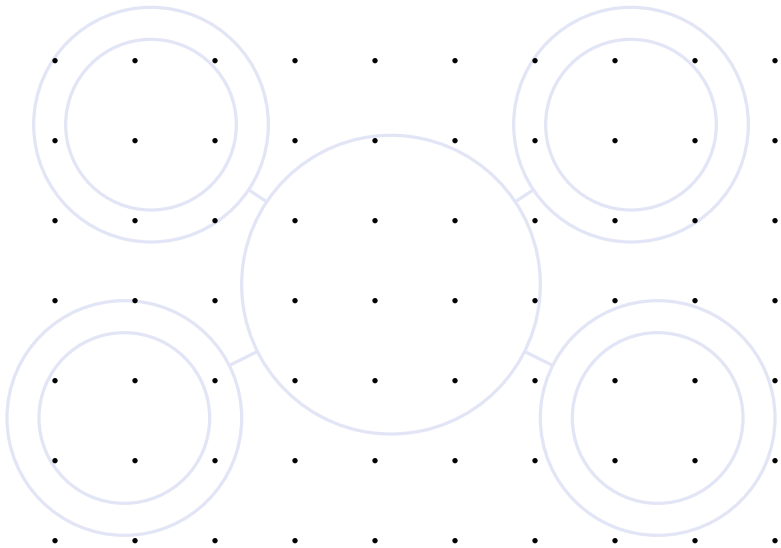
SAFE FLIGHT-PATH MAZE: GUIDE MATEO'S DRONE!

Draw a line from LAUNCH at top-left through open gates in order (1 -- 2 -- 3 -- 4) to LANDING PAD 'H'!



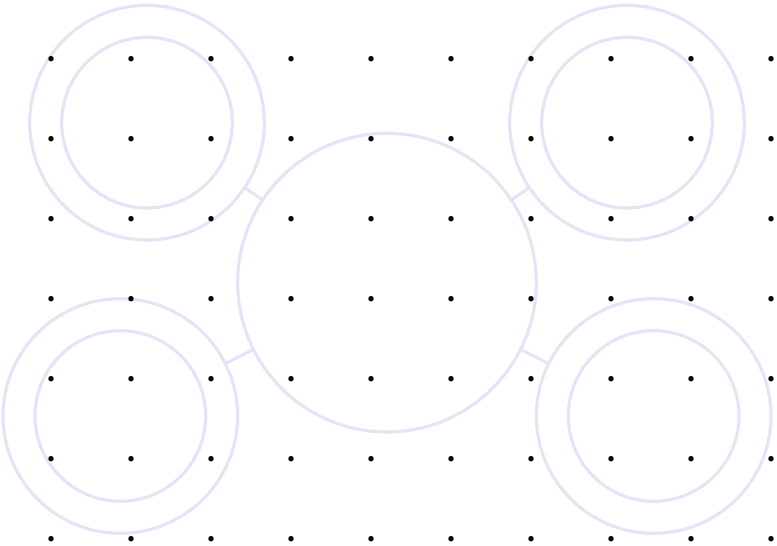
COLORING CHALLENGE: Color all 4 Gate Hoops different colors and draw arrows along your safe flight path!

BLEED-PREVENTION BACKING PAGE 16 - BOOK #052 - FLIGHT LOG & SKETCH PAD



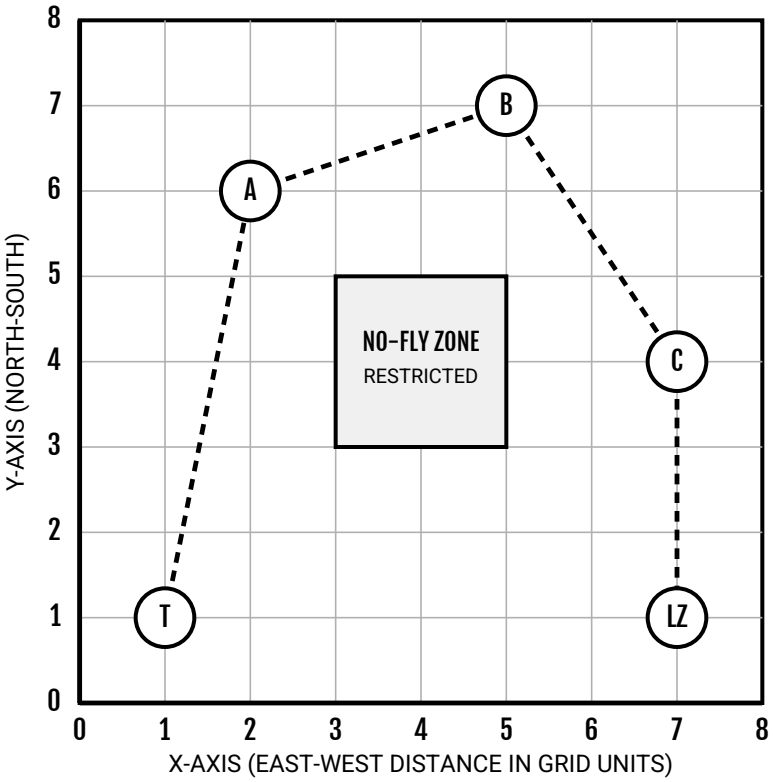


SAFE DRONE FLIGHT AT 3D PRINTING & MAKER WORKSHOP



COORDINATE FLIGHT PLAN: MATEO'S WAYPOINT NAVIGATION

Plot waypoints on the (X, Y) grid for Mateo's Dual Drone Rotor Addition!



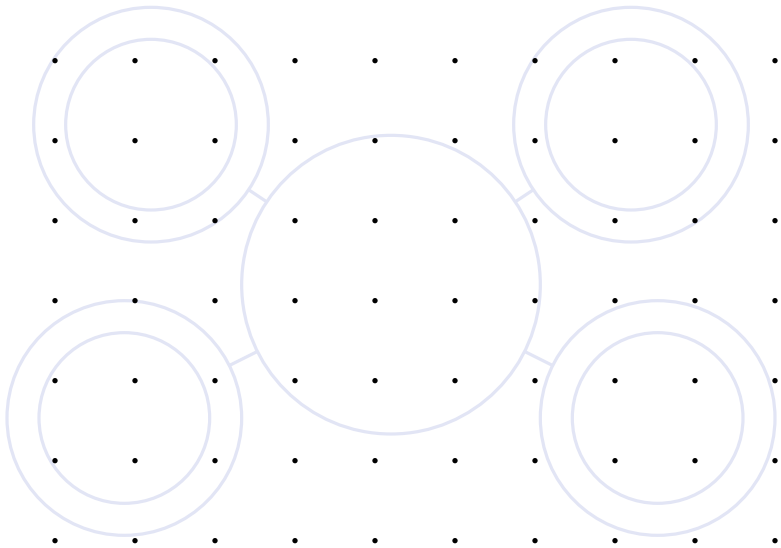
FLIGHT PLAN WAYPOINTS

WP T:	Takeoff Pad	(1 , 1)
WP A:	Mateo Survey	(2 , 6)
WP B:	Waypoint Dual Drone R	(5 , 7)
WP C:	Bridge Check	(7 , 4)
WP LZ:	Landing Zone	(7 , 1)

COORDINATE CHALLENGE:

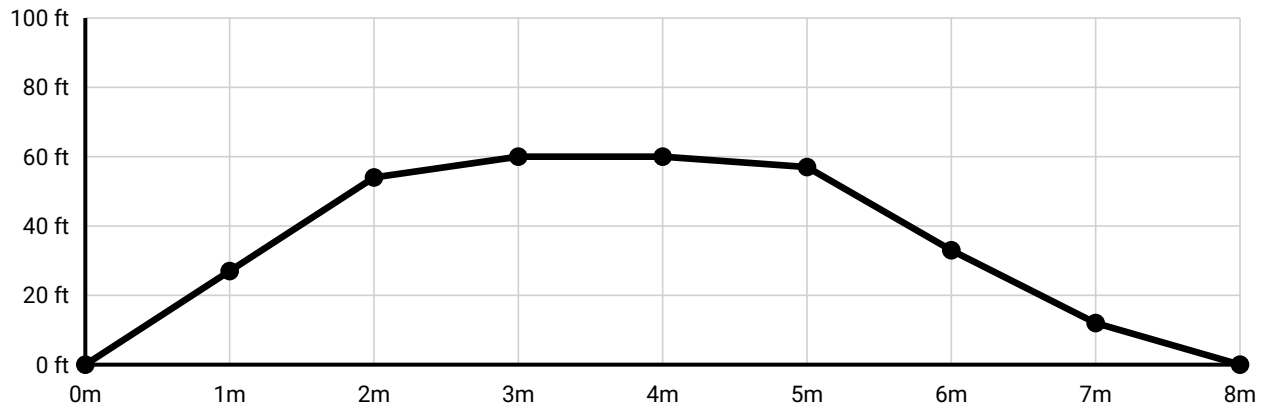
1. Waypoint B coordinates: (5 , 7)
2. Did flight avoid No-Fly Zone? [X] YES
3. Total flight legs flown: 4 legs

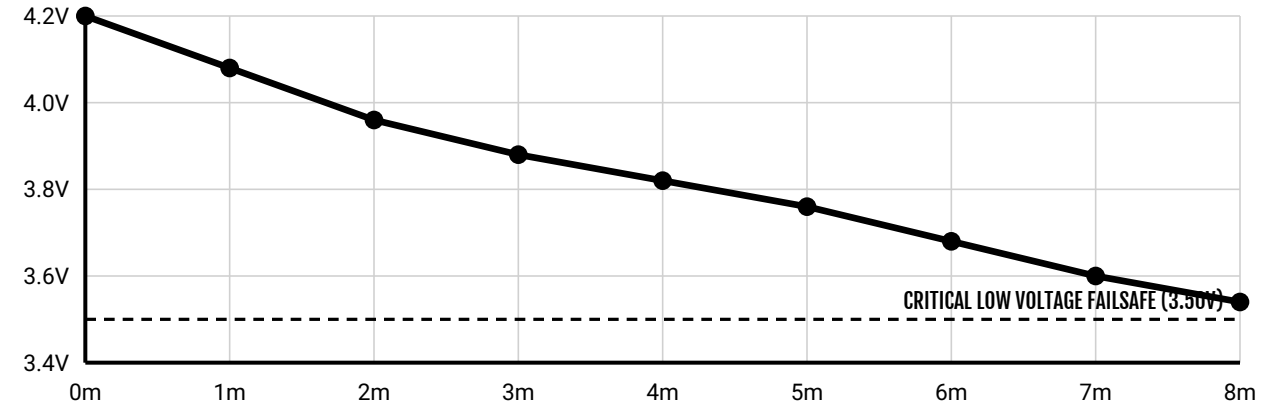
BLEED-PREVENTION BACKING PAGE 20 - BOOK #052 - FLIGHT LOG & SKETCH PAD



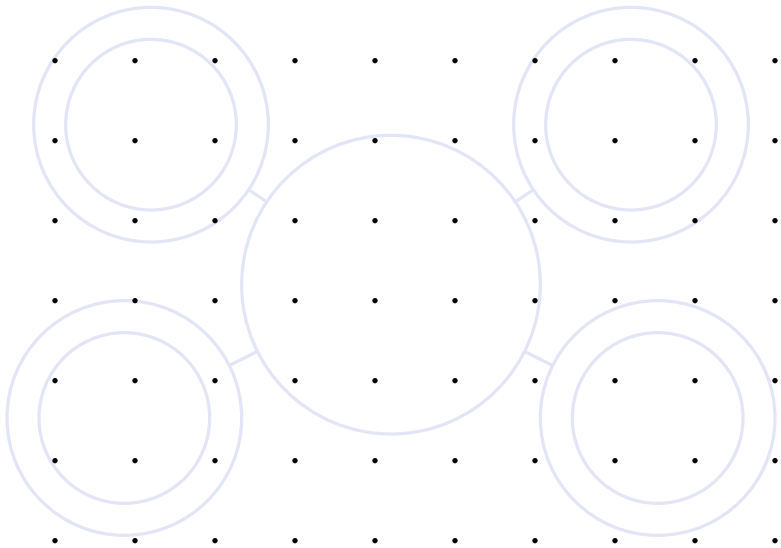
FLIGHT TELEMETRY: ALTITUDE & BATTERY DISCHARGE ENVELOPE

Analyze real flight telemetry data downloaded from the onboard BlackBox flight recorder.



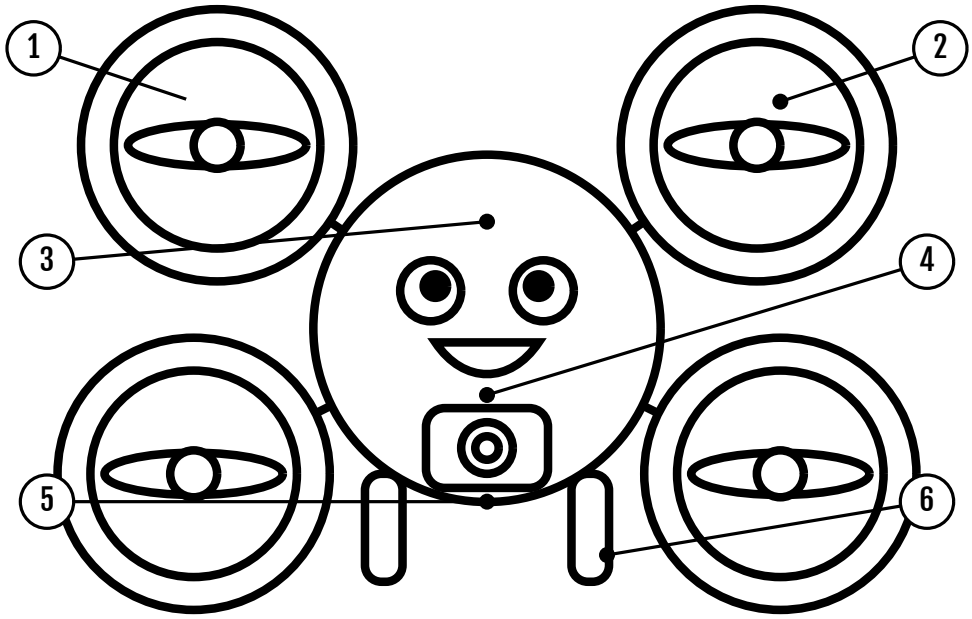


BLEED-PREVENTION BACKING PAGE 22 - BOOK #052 - FLIGHT LOG & SKETCH PAD



AEROSPACE BLUEPRINT: CUB-75 AVIONICS & AIRFRAME ANATOMY

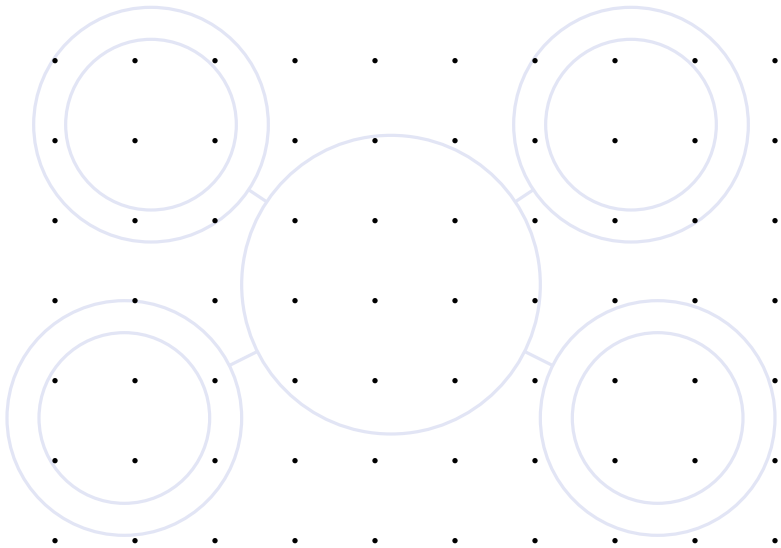
Study the mechanical airframe and avionics architecture of the Osito CUB-75 STEM training drone.



1	**Ducted Prop Guards**	Duct rings that shield propellers and provide 360-degree impact safety.
2	**Brushless DC Motors**	0802 19000KV motors that spin props at 30,000 RPM to produce vertical lift.
3	**Flight Controller (FC)**	Microprocessor & gyroscope that balances the drone 1,000 times every second.
4	**4-in-1 ESC Board**	Electronic Speed Controller regulating power sent from battery to each motor.
5	**Micro FPV Camera**	Ultra-wide lens sensor broadcasting real-time video to pilot FPV monitors.
6	**1S LiPo Battery**	3.8V High-Voltage Lithium Polymer rechargeable cell providing flight energy.

CUB-75 SPECS: Wheelbase: 75mm – AUW Weight: 39.5g – Motors: 0802 19000KV – Flight Time: 8.0 min

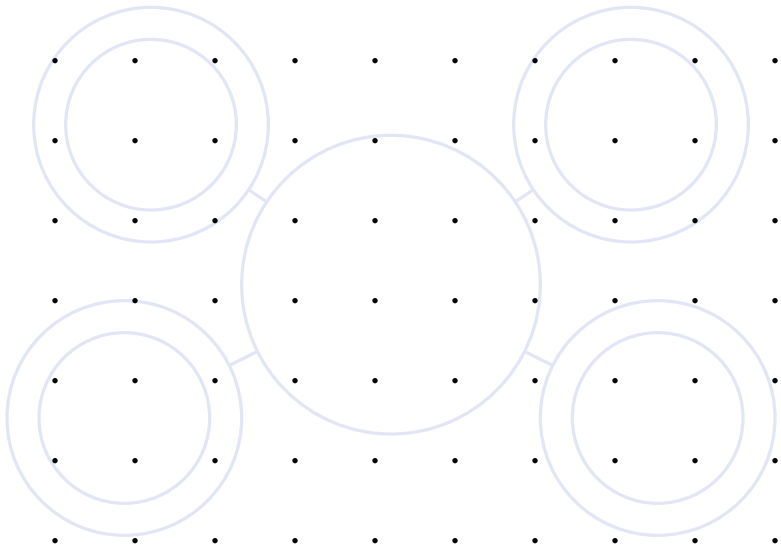
BLEED-PREVENTION BACKING PAGE 24 - BOOK #052 - FLIGHT LOG & SKETCH PAD



FLIGHT PHYSICS: THRUST-TO-WEIGHT & BATTERY ENDURANCE MATH

Apply real aeronautical engineering equations to calculate multirotor performance and endurance.

BLEED-PREVENTION BACKING PAGE 26 - BOOK #052 - FLIGHT LOG & SKETCH PAD



OFFICIAL OSITO JUNIOR PILOT FLIGHT MISSION LOG

RECORD YOUR ACTUAL FLIGHT MISSIONS AND COMPLETED STEM CHALLENGES

FLIGHT #	DATE	FLIGHT DURATION	MISSION OBJECTIVE	COACH SIGNATURE
Mission 1				
Mission 2				
Mission 3				

OSITO STEM PILOT CERTIFICATE OF ACHIEVEMENT

THIS CERTIFIES THAT JUNIOR PILOT

PILOT NAME HERE

HAS COMPLETED ALL STEM ACTIVITIES AND SAFETY PROTOCOLS IN

BOOK #052: DUAL DRONE ROTOR ADDITION

Core Aeronautical Skill: 4 Rotors Plus 4 Rotors Equals 8

AND IS OFFICIALLY RECOGNIZED AS A CERTIFIED JUNIOR OSITO DRONE PILOT!

JUNIOR PILOT SIGNATURE

COACH / PARENT SIGNATURE & DATE

BLEED-PREVENTION BACKING PAGE 28 - BOOK #052 - FLIGHT LOG & SKETCH PAD

